习题七

7-1

$$A_a=\begin{bmatrix}0&-1&0&0&1&1&1\\0&1&-1&1&0&-1&0\\-1&0&0&-1&0&0&-1\\1&0&1&0&-1&0&0\end{bmatrix}$$

$$A=\begin{bmatrix}0&-1&0&0&1&1&1\\0&1&-1&1&0&-1&0\\-1&0&0&-1&0&0&-1\end{bmatrix}$$

7-2 回路方向取连支方向

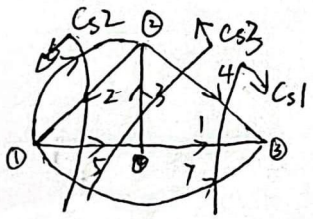


$$B_f=\begin{bmatrix}-1&0&1&1&0&0&0\\0&1&1&0&1&0&0\\0&1&0&0&0&1&0\\-1&0&0&0&0&0&1\end{bmatrix}$$

7-3

$$B_m=\begin{bmatrix}0&1&0&0&0&1&0\\0&-1&-1&0&-1&0&0\\-1&0&1&1&0&0&0\\1&0&0&0&1&0&-1\end{bmatrix}$$

7-4



$$Q_f=\begin{bmatrix}1&0&0&1&0&0&1\\0&1&0&0&-1&-1&-1\\0&0&1&-1&-1&0&-1\end{bmatrix}$$

7-5. 7-6 证明参见书本P183页
(虽然书本上只是举了个例子)

7-7

$B_t=-Q_l^T$ $B_f=[B_t\,|\,1]$

$Q_f=[1\,|\,Q_l]$

$$Q_f=\begin{bmatrix}1&0&0&1&-1&-1&0\\0&1&0&1&-1&-1&0\\0&0&1&1&0&0&-1\end{bmatrix}$$ (要转置)

7-8

$A\Upsilon A^T\dot{U}_n=A\Upsilon\dot{U}_S-A\dot{I}_S$ ① $Y_n=A\Upsilon A^T$

分别写出 $A.\dot{u}_S.\dot{i}_S.\Upsilon$.

$$A=\begin{bmatrix}1&0&-1&1&0&0\\0&-1&0&-1&1&0\\-1&1&0&0&0&-1\end{bmatrix}$$

$\dot{U}_S=[U_{S1},0,-U_{S3},0,0,0]^T$

$\dot{I}_S=[0,I_{S2},0,0,0,0]^T$

$\Upsilon=diag[\frac{1}{R_1},\frac{1}{R_2},\frac{1}{R_3},\frac{1}{R_4},\frac{1}{R_5},\frac{1}{R_6}]$

$Y_n=A\Upsilon A^T=[\quad]$

代入①即可

7-9

$B_f z B_f^T\dot{I}_l=B_f z\dot{I}_S-B_f\dot{U}_S$

分别写出 $B_f.z.\dot{i}_S.\dot{u}_S$

$$B_f=\begin{bmatrix}1&0&0&-1&-1&-1\\0&1&0&0&1&1\\0&0&1&1&1&0\end{bmatrix}$$

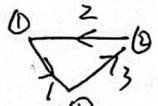
$z=diag[R_1,R_2,R_3,R_4,R_5,R_6]$

$\dot{U}_S=[U_{S1},0,-U_{S3},0,0,0]^T$

$\dot{I}_S=[0,I_{S2},0,0,0,0]^T$

代入即可

7-10



$$\Upsilon=\begin{bmatrix}\frac{1}{R_1}+\frac{1}{j\omega L_1}&0&0\\0&\frac{1}{j\omega L_2}+\frac{1}{R_4+\frac{1}{j\omega C_4}}&0\\g_m&0&\frac{1}{R_3}\end{bmatrix}$$

$Y_n=A\Upsilon A^T$ ① $A=\begin{bmatrix}1&-1&0\\0&1&-1\end{bmatrix}$

代入①可算得

$$Y_n=A\Upsilon A^T=\begin{bmatrix}\frac{1}{R_1}-j\frac{1}{\omega L_1}+\frac{1}{j\omega L_2}+\frac{1}{R_4-j\frac{1}{\omega C_4}} & j\frac{1}{\omega L_2}-\frac{1}{R_4-j\frac{1}{\omega C_4}}\\ -g_m+j\frac{1}{\omega L_2}-\frac{1}{R_4-j\frac{1}{\omega C_4}} & \frac{1}{R_3}-j\frac{1}{\omega L_2}+\frac{1}{R_4-j\frac{1}{\omega C_4}}\end{bmatrix}$$

用 $A\Upsilon A^T\dot{U}_n=A\Upsilon\dot{U}_S-A\dot{I}_S$

列出节点电压方程

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7-11

$$Y = \mathrm{diag}\left[jwC_1 + \frac{1}{R_1 + jwL_1},\ jwL_2 + \frac{1}{\frac{1}{R_2} + jwC_2}\right]$$

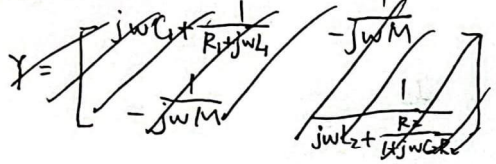
$$A = [1, 1] \qquad U_S = [0, U_S]^T$$

$$Y_n = AYA^T = \begin{bmatrix} -j\frac{1}{wL_2} + jwC_1 + \frac{1}{R_1 + jwL_1} & j\frac{1}{wL_2} \\ j\frac{1}{wL_2} & -j\frac{1}{wL_2} + jwC_2 + \frac{1}{R_2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{R_1 + jwL_1} + jwC_1 + \frac{1}{jwL_2} & -\frac{1}{jwL_2} \\ -\frac{1}{jwL_2} & \frac{1}{R_2} + jwC_2 + \frac{1}{jwL_2} \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{U_S}{R_2} \end{bmatrix}$$

$$AYA^T U_n = AYU_S - AI_S$$

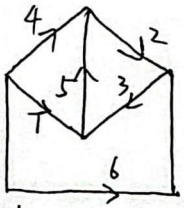
7-12



互感相当于电流控制电压源

$$Z = \begin{bmatrix} \frac{1}{jwC_1 + \frac{1}{R_1 + jwL_1}} & -jwM \\ -jwM & jwL_2 + \frac{R_2}{1 + jwC_2R_2} \end{bmatrix} \qquad Y = Z^{-1}$$

7-13



即选1.2.3为树支

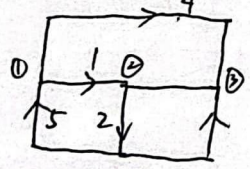
$$B_f = \begin{bmatrix} -1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$Z = \begin{bmatrix} jwL_1 & jwM & & & & \\ jwM & jwL_2 & & & & \\ & & \frac{1}{jwC_3} & & & \\ & & & R_4 & & \\ & & & & R_5 & \\ & & & & & R_6 \end{bmatrix}$$

$$Z_n = B_f Z B_f^T$$

$$B_f Z B_f^T I_1 = B_f Z I_S - B_f U_S$$

7-14



$$Z = \begin{bmatrix} jwL_1 & jwM_{12} & & & & \\ jwM_{12} & jwL_2 & -jwM_{23} & & & \\ & -jwM_{23} & jwL_3 & & & \\ & & & R_4 & & \\ & & & & R_5 & \\ & & & & & \frac{1}{jwC_6} \end{bmatrix}$$

$$B_f = \begin{bmatrix} -1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Z_n = B_f Z B_f^T \qquad Z_n I_1 = B_f Z I_S - B_f U_S$$

7-15

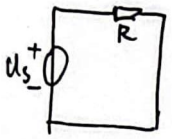
$$\begin{bmatrix} R_1 + R_2 + R_4 & -R_4 & -R_2 \\ -R_4 & R_3 + R_4 + R_5 & -R_5 \\ -R_2 & -R_5 & R_2 + R_5 + R_6 \end{bmatrix} \begin{bmatrix} I_{L1} \\ I_{L2} \\ I_{L3} \end{bmatrix} = \begin{bmatrix} -U_{S1} + \\ I_S R_2 \\ U_{S3} \\ -R_2 I_{S2} \end{bmatrix}$$

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8-1

$u_C(0^+)=u_C(0^-)=0V$

$t=0^+$ 时等效电路为



$I_R(0^+)=I_C(0^+)=\frac{u_S}{R}=0.1A$

$\frac{di_C(0^+)}{dt}=\frac{-1}{R}\frac{du_C(0^+)}{dt}=\frac{-1}{RC}I_{(0^+)}=-100A/s$

⇓

第二类非独立初值

$u_S=Ri_C+u_C \xrightarrow{求导} 0=R\frac{di_C}{dt}+\frac{du_C}{dt}$

$\Rightarrow \frac{di_C}{dt}=-\frac{1}{R}\frac{du_C}{dt}$

而 $\frac{di_C}{dt}=-\frac{1}{R}\frac{du_C}{dt}=-\frac{1}{RC}\cdot i_C$

$\frac{d^2i_C}{dt^2}=-\frac{1}{RC}\cdot\frac{di_C}{dt}$

即 $\frac{d^2i_C(0^+)}{dt}=-\frac{1}{RC}\cdot\frac{di_C(0^+)}{dt}=10^5A/s^2$

8-2

$I_L(0^+)=I_L(0^-)=0A$

闭合后 $t=0^+$ 时 $u_L(0^+)=u_S=100V$

$u_L(0^+)=L\frac{di_L(0^+)}{dt}\Rightarrow\frac{di_L(0^+)}{dt}=\frac{1}{L}u_L(0^+)=100A/s$

$\frac{d^2I_L(0^+)}{dt^2}=\frac{1}{L}\frac{du_L(0^+)}{dt}$ ①

下面求 $\frac{du_L(0^+)}{dt}$

$t>0$ 时 $u_S=i_L\cdot R+u_L \xrightarrow{求导} \frac{du_L}{dt}=-R\cdot\frac{di_L}{dt}$

$\therefore \frac{du_L(0^+)}{dt}=-10\times100=-1000V/s$

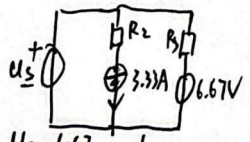
代入①式得 $\frac{d^2I_L(0^+)}{dt^2}=-1000A/s^2$

8-3

$I_1(0^+)=\frac{u_S}{R_1+R_2}=3.3A$

$u_C(0^+)=u_C(0^-)=u_S\cdot\frac{R_2}{R_1+R_2}=66.7V$

$t=0^+$ 时



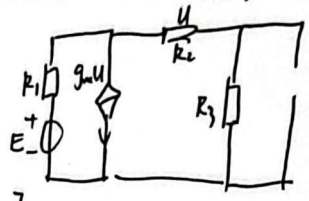
$i_2(0^+)=\frac{u_S-66.7}{R_3}=6.67A$

即 $i_1(0^+)=\frac{10}{3}A$ $i_2(0^+)=\frac{5}{3}A$

8-4

$i(0^+)=i(0^-)=0A$ $u_C(0^+)=u_C(0^-)=0V$

$t=0^+$ 时



$U=R_2I_2$

$\left.\begin{aligned}E&=R_1I_1+(R_2+R_3)I_2\\I_1&=g_mR_2I_2+I_2\end{aligned}\right\}\Rightarrow\begin{cases}I_1=1.5A\\I_2=0.5A\end{cases}$

$u_L(0^+)=R_3I_2=1V=L\frac{di_L(0^+)}{dt}$

$\Rightarrow\frac{di}{dt}(0^+)=\frac{u_L(0^+)}{L}=1A/s$

8-5 为奇异电路 (×) 注意观察发现闭合前并不是奇异电路

~~$I_S(0^+)=I_{L_1}(0^+)+I_{L_2}(0^+)$~~ ~~闭合后为奇异~~ 因为有在一个R，也属于割集的

~~$L_1I_{L_1}(0^+)+L_2I_{L_2}(0^+)=L_1I_{L_1}(0^-)-L_2I_{L_2}(0^-)$~~ 一部分

开关闭合瞬间，满足换路定则

即 $i_2(0^+)=i_2(0^-)=2A$

而 $t\to\infty$ 时，显然 R 被短路，这时可以用奇异电路的思路求解，即运用磁链守恒

$L_1i_1(\infty)-L_2i_2(\infty)=L_1i_1(0^-)-L_2i_2(0^-)$

$I_S=i_1(\infty)+i_2(\infty)$

$\Rightarrow i_1(\infty)=i_2(\infty)=3A$

即 $i_2(0^+)=2A$ $i_2(\infty)=3A$

8-6 闭合后为奇异电路

故 $u_{S_1}=u_{C_1}+u_{C_2}$

电荷守恒 $C_1u_{C_1(0^+)}-C_2u_{C_2}(0^+)=C_1u_{C_1}(0^-)-C_2u_{C_2}(0^-)$

闭合前 $u_{C_1}(0^-)=u_S\cdot\frac{R_1}{R_1+R_2}=3V$ $u_{C_2}(0^-)=0V$

即 $u_{C_1}(0^+)-2u_{C_2}(0^+)=3$

$u_{C_1}(0^+)+u_{C_2}(0^+)=9$

$\Rightarrow u_{C_1}(0^+)=7V$ $u_{C_2}(0^+)=2V$

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8-7 (本题求全响应)

$U_1 = -R_1 i_1$

$i_L(0^+) = i_L(0^-) = 0A$

$t \to \infty$ 时



$i_{Lp}(t) = i_L(\infty) = i_1(\infty) + g_m U_1$

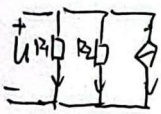
$i_1 = \frac{E}{R_1} = 6A \quad -g_m U_1 = -g_m(-I_1 R_1) = 12A$

$i_{Lp}(t) = 18A$

$i_L(t) = i_{Lp}(t) + [i_L(0^+) - i_{Lp}(0^+)]e^{-\frac{t}{\tau}}$

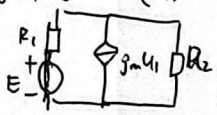
$\tau = \frac{L}{R_{eq}} = 4S$

$G_{eq} = \frac{1}{R_1} + \frac{1}{R_2} + g_m = 4S \quad R_{eq} = 0.25\Omega$



$i_L(t) = 18 - 18e^{-0.25t}(A)$

$t = 0^+$ 时



$E = R_1 I_1 + (I_1 + g_m I_1 R_1) R_2$

$\Rightarrow I_1 = 1.5A \quad g_m U_1 = -g_m I_1 R_1 = -3A$

$t \to \infty$ 时 $g_m U_1 = -g_m I_1 R_1 = -12A$

即 $g_m U_{1p}(t) = g_m U_1(\infty) = -12A$

$g_m U_1(t) = -12 + 9e^{-0.25t}(A)$

8-8

$t = 0^-$ 时 $I_{L_1} = \frac{U_S}{10 // 30} = \frac{4}{3}A$

$I_{L_1}(0^+) = I_{L_1} = \frac{4}{3}A$

$t \to \infty$ 时

$R_{总} = 10 // (10 + 20 // 20) = \frac{20}{3}\Omega$

$I_{总} = 1.5A$

$I_{L_1} = I_{总} \times \frac{20}{10+20} + I_{总} \times \frac{10}{10+20} \times \frac{1}{2}$

$= 1.25A$

~~$\tau = R_{总} \cdot$~~ $R_{eq} = 10 // (20 + 10 // 20) = \frac{80}{11}\Omega$

~~$\tau = R_{eq} \cdot L$~~ $\tau = \frac{L_1}{R_{eq}} = \frac{33}{1600}S$

$I_{L_1}(t) = 1.25 + \frac{1}{12}e^{-\frac{1600}{33}t}(A)$

$U_2(t) = 1 \cdot \frac{di_{L_1}(t)}{dt} = -0.2e^{-\frac{1600}{33}t}$

8-9 (注意：非正弦源/直流源不能用三要素法)

$i(t) = 2t \quad (0 \le t \le 1)$

KVL可得 $U_L - [i(t) - i_L(t)] \cdot R = 0$

$L\frac{di_L(t)}{dt} + Ri_L(t) = 2t$

$0.5\frac{di_L}{dt} + i_L = 2t$

$0 \le t \le 1$ 时 $i_L(t) = 2t - 1 + Ce^{-2t}$

由 $i_L(0) = 0 \Rightarrow C = 1$

即 $i_L(t) = 2t - 1 + e^{-2t}(A) \quad U_L(t) = L\frac{di_L}{dt} = 1 - e^{-2t}(V)$

② $t > 1$ 时

$\frac{di_L}{dt} + 2i_L = 0 \Rightarrow i_L(t) = ke^{-2(t-1)}$

$i_L(1) = 1 + e^{-2} \Rightarrow k = 1.135/e^2 \quad i_L(t) = 1.135e^{-2(t-1)}$

$U_L(t) = L\frac{di_L}{dt} = -1.135e^{-2(t-1)}V \quad = 1.135e^{-2t}A$

$$U_L(t) = \begin{cases} 1 - e^{-2t}(V) & 0 \le t \le 1 \\ -1.135e^{-2(t-1)}(V) & t \ge 1 \end{cases}$$

8-10

$\frac{1}{2}CU_0^2 = 5J \Rightarrow U_C(0^+) = U_C(0^-) = 500V$

$U_C(\infty) = U_S = 0V$



$U_C(t) = 500e^{-\frac{t}{\tau}} = 500 \times \frac{\sqrt{2}}{2}$ (能量减半，电压减到$\frac{\sqrt{2}}{2}$)

其中 $t_1 = 5.56 \times 10^{-3}S$

$\Rightarrow \tau = 0.016S = RC \Rightarrow R = 401.07\Omega$

8-11

$t < 0$ 时 $\dot{U}_S = 1\angle -45°$

$U_C(0^+) = U_C(0^-) = 2V \quad i_C(0^+) = \frac{U_S - U_C(0^+)}{R} = -0.3A$

$t \to \infty$ 时 $\dot{U}_S = 1\angle -45°$

$\dot{I} = \frac{\dot{U}_S}{R_{并联}} = 0.053\angle 12.87° \quad \frac{1}{j\omega C} = -j5.92$

$\dot{U}_C = \dot{I} \cdot \frac{1}{j\omega C} = 0.847\angle -77.13°$

即 $i_{Cp}(t) = i_C(\infty) = 0.053\sqrt{2}\sin(314t + 12.87°)A$

$U_{Cp}(t) = U_C(\infty) = 0.847\sqrt{2}\sin(314t - 77.13°)V$

根据三要素法： $\tau = RC = 2 \times 10^{-3}S$

$i_{Cp}(0) = 0.0167A \quad U_{Cp}(0) = -1.168V$

$i_C(t) = 0.053\sqrt{2}\sin(314t + 12.87°) + (-0.3167e^{-500t})(A)$

$U_C(t) = 0.847\sqrt{2}\sin(314t - 77.13°) + 3.168e^{-500t}(V)$

8-12

$t = 0^-$ 时 由KCL可知 $1 \times (1 // 0.5i) = 2i \Rightarrow i = 0.4A$

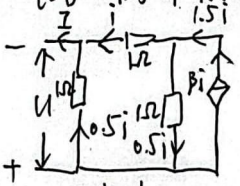
$U_t(0^+) = U_t(0^-) = Ri = 0.4倍V$

$t = \infty$ 时 由KCL可知 $1 \times (1 + 0.5i) = 2i \Rightarrow i = \frac{2}{3}A$

$U_{tp}(t) = U_t(\infty) = \frac{2}{3}V$

下面算 $\tau = R_{eq}$

电容左侧作戴维南等效



$\beta = 1.5$ 易知 $I = 1.5i$

$U = 0.5i \cdot R = 0.5i$

$R_{eq} = \frac{U}{I} = \frac{1}{3}\Omega$

$\tau = R_{eq} \cdot C = \frac{1}{3}S$

由三要素法

$\therefore U_t = \frac{2}{3} - \frac{4}{15}e^{-3t}(V)$

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8-13

$u_1(0^+)=u_1(0^-)=8V$

$u_2(0^+)=u_2(0^-)=2V$

$t=\infty$ 时 利用电荷守恒.

此时R上无电流通过. 即$u_1(\infty)=u_2(\infty)$

$C_1u_1(0^-)+C_2u_2(0^-)=(C_1+C_2)u_1(\infty)$

$u_1(\infty)=u_2(\infty)=6V$

$\tau=\frac{C_1C_2}{C_1+C_2}\cdot R=\frac{1}{3}s$

$\therefore u_1(t)=6+2e^{-3t}(V)\ u_2(t)=6-4e^{-3t}(V)$

$i(t)=\frac{u_1(t)-u_2(t)}{R}=6e^{-3t}(A)$

$W_1=\frac{1}{2}C_1u_1^2(\infty)=18J\quad W_2=\frac{1}{2}C_2u_2^2(\infty)=9J$

8-14 断开后. 左右两部分相对保持独立

$t=0^-$时 $u_S=i_1R_1+ri_1\Rightarrow i_1=0.2A$

$u_{C2}(0^-)=4V\quad u_{C1}(0^-)=8V$

$t=0^+$时 $u_{C2}(0^+)=u_{C2}(0^-)=4V$

$u_{C1}(0^+)=u_{C1}(0^-)=8V$

左侧 $t=+\infty$时 $u_{C1}(\infty)=12V$

由三要素法 ~~$u_{C1}(t)=8+4e^{-}$~~

$u_{C1}(t)=12-4e^{-\frac{t}{2}}(V)\ u_{0t}(t)=u_{C1}(t)-ri_1(t)$

$=12-12e^{-\frac{1}{2}t}(V)$

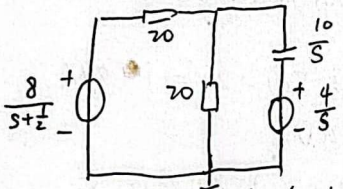
($\tau_1=R_1C_1=2S$)

观察到右侧由电压源ri_1供压

而$i_1=0.2e^{-\frac{t}{2}}A\quad 40i_1=8e^{-\frac{t}{2}}A$

为指数型电压源. 不能用三要素法.

故用第九章的方法给元件建模计算



用节点电压法 $(\frac{1}{20}+\frac{1}{20}+\frac{10}{s})U_0=\frac{8/20}{\frac{1}{s}+\frac{1}{2}}+\frac{\frac{4}{s}}{\frac{10}{s}}$

$\Rightarrow U_{C2}(s)=\frac{8}{s+\frac{1}{2}}+\frac{-4}{s+1}$

$\Rightarrow u_{C2}(t)=8e^{-\frac{t}{2}}-4e^{-t}(V)$

(也可以用微分方程法求解.)

8-15

与上题类似. 左右电路相对独立

先求左边

$i_L(\infty)=i_L(\frac{t}{\infty})=U_S\cdot\frac{R}{R+R//R}\cdot\frac{1}{2}=1A$

$i_L(0^+)=i_L(0^-)=0A$

由三要素法

$i_L(t)=1-e^{-\frac{t}{\tau}}\quad \tau=\frac{L}{Req}=\frac{L}{R+R//R}=0.1s$

故$i_L(t)=1-e^{-10t}(A)$

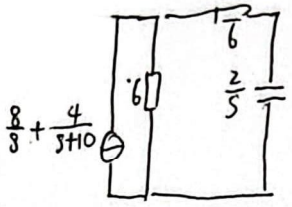
(续8-15)

求$i(t)$

$Ri(t)=Ri_L(t)+L\frac{di_L}{dt}\Rightarrow 1+0.5e^{-10t}=i(t)$

即$i(t)=1+0.5e^{-10t}(A)\quad \alpha_i=8+4e^{-10t}(A)$

对右侧元件建模



$U_C(s)=(\frac{8}{s}+\frac{4}{s+10})\cdot\frac{6\cdot\frac{2}{5}}{6+\frac{2}{5}+6}$

$=48(\frac{1}{s}-\frac{1}{s+6})+\frac{24}{59}(\frac{1}{s+6}-\frac{1}{s+10})$

$=48\frac{1}{s}-47.59\frac{1}{s+6}-0.407\frac{1}{s+10}$

由拉氏反变换 $u_C(t)=48-47.59e^{-6t}-0.407e^{-10t}(V)$

(同可用微分方程法求解. 由于本人常微分方程学艺不精. 暂不列出)

8-16

① 开关闭合前 $0\le t<1$时

$t\to 0$时$u_S(t)=(R_1+R_2)i_L+L\frac{di_L}{dt}$

$\delta(t)=2i_L+\frac{di_L}{dt}\Rightarrow\int_{0^-}^{0^+}\delta(t)dt=\int_{0^-}^{0^+}\frac{di_L}{dt}dt$

$\because i_L(0^-)=0A\quad i_L(0^+)-i_L(0^-)=1A$

$\therefore i_L(0^+)=1A\quad i_L(\infty)=0A\quad \tau=\frac{L}{Req}=\frac{L}{R_1+R_2}=0.5s$

由三要素法 $i_L(t)=e^{-2t}(A)$

② 开关闭合后 $1\le t$时

$i_L(1^+)=i_L(1^-)=e^{-2}A$

$i_L(\infty)=\frac{1}{3}A\quad \tau=\frac{L}{Req}=\frac{L}{1+1//1}=\frac{2}{3}s$

$\therefore i_L(t)=\frac{1}{3}+(e^{-2}-\frac{1}{3})e^{-\frac{3}{2}(t-1)}\cdot\varepsilon(t-1)(A)$

即 $i_L(t)=\begin{cases}e^{-2t}A & 0\le t<1\\ \frac{1}{3}+(e^{-2}-\frac{1}{3})e^{-\frac{3}{2}(t-1)}(A) & t\ge 1\end{cases}$

8-17

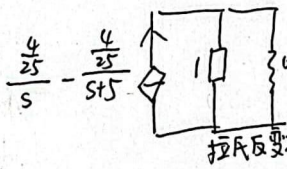
左侧 $u_C(0^+)=u_C(0^-)=0V$

$u_C(\infty)=1\times\frac{4}{5}=\frac{4}{5}V\quad \tau=Req\cdot C=0.25\times(4//1)=0.25s$

三要素 $u_C(t)=\frac{4}{5}-\frac{4}{5}e^{-5t}(V)$

$0.2u_C(t)=\frac{4}{25}-\frac{4}{25}e^{-5t}(A)$

对右侧电路建模



$I_L(s)=[\frac{\frac{4}{25}}{s}-\frac{\frac{4}{25}}{s+5}]\cdot\frac{1}{1+0.5s}$

$=\frac{4}{25}(\frac{1}{s}-\frac{1}{s+2})-\frac{8}{75}(\frac{1}{s+2}-\frac{1}{s+5})$

拉氏反变换 $I_L(t)=\frac{4}{25}-\frac{4}{15}e^{-2t}+\frac{8}{75}e^{-5t}(A)$

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8-18

运用对偶定理：电感L无穷状态对应电容C零状态
电容C无穷状态对应电感L零状态

由零状态响应易知

电感L作用时 $u_0(\infty)=3V \quad u_0(0^+)=-4.5V$

$\tau=\frac{L}{Req}=\frac{1}{10}s \Rightarrow Req=9\Omega$

故电容C作用时 $u_0'(\infty)=-4.5V \quad u_0'(0^+)=3V$

$\tau'=Req\cdot C=\frac{1}{10}s$

即 换成电容C后 $u_0'(t)=-4.5+7.5e^{-10t}(V)$

8-19 同样运用对偶定理

分析图a

$\tau=Req\cdot C=\frac{1}{25}s \Rightarrow Req=4000\Omega$

$i_C(0^+)=\frac{1}{6}mA \quad i_C(\infty)=0A$

运用对偶定理

若电路仍采用1(t)电源，即将图b中δ(t)换成1(t)

则 $i_L(0^+)=0A \quad i_L(\infty)=\frac{1}{6}mA$

$\tau'=\frac{L}{Req}=\frac{1}{1000}s$

故 $i_L'(t)=\frac{1}{6}-\frac{1}{6}e^{-1000t}(mA)$

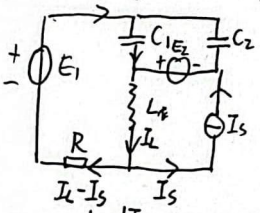
$u'_L(t)=L\frac{di_L'(t)}{dt}=\frac{2000}{3}e^{-1000t}(mV)$

$=\frac{2}{3}e^{-1000t}\cdot 1(t)(V)$

$u_L(t)=\frac{du'_{L(t)}}{dt}=\frac{2}{3}e^{-1000t}\delta(t)-\frac{2000}{3}e^{-1000t}\cdot 1(t)(V)$

(换成δ(t)电源) $=\frac{2}{3}\delta(t)-\frac{2000}{3}e^{-1000t}\cdot 1(t)(V)$

8-20



$E_1=u_{C1}+\frac{L}{R}\frac{dI_L}{dt}+(I_L-I_S)R$ ①

$u_{C2}=u_{C1}+E_2$ ②

$C_1\frac{du_{C1}}{dt}+C_2\frac{du_{C2}}{dt}=I_L-I_S$ ③

将①②代入③

$C_1(-L\frac{d^2I_L}{dt^2}-R\frac{dI_L}{dt})+C_2(-L\frac{d^2i_L}{dt^2}-R\frac{di_L}{dt})=I_L-I_S$

$(C_1L+C_2L)\frac{d^2i_L}{dt^2}+(C_1R+C_2R)\frac{di_L}{dt}+I_L=I_S$

代入参数

$9\times10^{-8}\frac{d^2i_L}{dt^2}+3\times10^{-4}\frac{di_L}{dt}+i_L=I_S$

暂态分量为 $9\times10^{-8}\frac{d^2i_L}{dt^2}+3\times10^{-4}\frac{di_L}{dt}+i_L=0$ 的解

特征方程为 $9\times10^{-8}s^2+3\times10^{-4}s+1=0$

$s_1=\frac{1}{3}\times10^4\angle120^\circ \quad s_2=\frac{1}{3}\times10^4\angle-120^\circ$

$i_L=Ae^{-\frac{1}{6}\times10^4t}\sin(\frac{\sqrt{3}}{6}\times10^4t+\theta)$

8-21

$t>0$时

$\begin{cases} i_L=\frac{u_C}{50}+C\frac{du_C}{dt} & ① \\ u_L+u_C=0 & ② \quad u_L=L\frac{di_L}{dt} \end{cases}$

$\Rightarrow 10^{-6}\frac{d^2i_L}{dt^2}+\frac{1}{400}\frac{di_L}{dt}+i_L=0$

特征方程为 $10^{-6}s^2+\frac{1}{400}s+1=0$

$s_1=-500, \quad s_2=-2000$

即 $i_L(t)=C_1e^{-500t}+C_2e^{-2000t}$

初值条件：

$i_L(0^-)=1\times\frac{75}{50+75}=0.6A=i_L(0^+)$

$u_C(0^+)=u_C(0^-)=0.6\times50=3V$

由②知 $-500C_1+(-2000)C_2=-24$

$C_1+C_2=0.6$

$\Rightarrow \begin{cases} C_1=0.784 \\ C_2=-0.184 \end{cases}$

$i_C(t)=0.784e^{-500t}-0.184e^{-2000t}\varepsilon(t)A$

8-22

分析可得

$2i_L+L\frac{di_L}{dt}=u_C \Rightarrow \frac{d^2i_L}{dt^2}+2\frac{di_L}{dt}+1=0$

$C\frac{du_C}{dt}=-i_L \Rightarrow i_L(t)=(A_1+A_2t)e^{-t}(A)$

而 $u_C(0^+)=u_C(0^-)=1V$

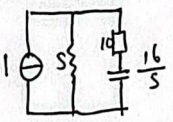
$I_L(0^+)=I_L(0^-)=1A \quad \frac{di_L}{dt}(0^-)=-1A/s$

代入得 $A_1=1 \quad A_2=0$

$i_L(t)=e^{-t}A \quad u_C(t)=e^{-t}V$

8-23

① 对元件建模



$I_L(s)=1\times\frac{10+\frac{16}{5}}{5+10+\frac{11}{5}}=\frac{10s+16}{s^2+10s+16}$

$=\frac{10s+20-4}{s^2+10s+16}=\frac{10}{s+8}-\frac{2}{3}(\frac{1}{s+2}-\frac{1}{s+8})$

$=\frac{32}{3}\cdot\frac{1}{s+8}-\frac{2}{3}\frac{1}{s+2}$

$i_L(t)=\frac{32}{3}e^{-8t}-\frac{2}{3}e^{-2t}(A)$

$i_L(0^+)=10A$

$U_C(s)=1\times\frac{s}{s+10+\frac{11}{5}}\cdot\frac{16}{5}=\frac{64}{3}\cdot\frac{1}{s+8}-\frac{16}{3}\cdot\frac{1}{s+2}$

$u_C(t)=\frac{64}{3}e^{-8t}-\frac{16}{3}e^{-2t}(V)$

$u_C(0^+)=16V$

② 把电感看成断路，把电容看成短路（冲击响应时

$i_C(t)=\delta(t) \quad u_L(t)=10\delta(t)$

$u_C(0^+)=u_C(0^-)+\frac{1}{C}\int_{0^-}^{0^+}i_C(t)dt=16V$

$i_L(0^+)=i_L(0^-)+\frac{1}{L}\int_{0^-}^{0^+}u_L(t)dt=10A$

$L\frac{di_L}{dt}=10i_C+\frac{1}{C}\int i_Cdt \Rightarrow L\frac{d^2i_L}{dt^2}+10\frac{di_L}{dt}+\frac{1}{C}i_L=0$

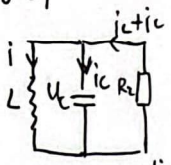
$\Rightarrow i_L(t)=A_1e^{-2t}+A_2e^{-8t}$

$i_L(0^+)=10A \quad \frac{di_L}{dt}=-84V/s$

$i_L(t)=-\frac{2}{3}e^{-2t}+\frac{32}{3}e^{-8t}A.$

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8-24



$u_C=R_1 i_L+L\frac{di_L}{dt}$

$u_C+(i_L+C\frac{du_C}{dt})R_2=0$

$\Rightarrow 2\frac{d^2i_L}{dt^2}+3\frac{di_L}{dt}+3i_L=0$

特征方程为 $2s^2+3s+3=0$

$\Rightarrow s_1=\frac{-3+\sqrt{15}j}{4}$ $s_2=\frac{-3-j\sqrt{15}}{4}$

故 $i_L(t)=Ae^{-\frac{3}{4}t}\sin(\frac{\sqrt{15}}{4}t+\theta)$

$i_L(0+)=-0.5A$ $u_L(0+)=1V$

$\frac{du_C}{dt}|_{0+}=0.5A/s$

代入初值得 $A=\frac{2\sqrt{15}}{15}$ $\theta=0$

$i_L(t)=\frac{2\sqrt{15}}{15}e^{-\frac{3}{4}t}\sin\frac{\sqrt{15}}{4}t$

8-25 (使特征方程 $\Delta=0$)

同样有 $u_C=i_L+\frac{2di_L}{dt}$

$i_L+2\frac{di_L}{dt}+R_2i_L+0.5R_2\frac{du_C}{dt}=0$

代入得 $R_2\frac{d^2i_L}{dt^2}+(0.5R_2+2)\frac{di_L}{dt}+(1+R_2)i_L=0$

$\Delta=(0.5R_2+2)^2-4R_2(1+R_2)=0\Rightarrow R_2=0.8\Omega$

8-26

$i_L(0+)=i_L(0^-)=6\times\frac{1}{10+2}=0.5A$

$u_C(0+)=u_C(0^-)=R_2i_L(0+)=1V$

当 $t>0$ 时

$\begin{cases}12=L\frac{di_L}{dt}+R_1i_L+u_C①\\ u_C=R_2(i_L-C\frac{du_C}{dt})②\end{cases}$

$u_C=12-2\frac{di_L}{dt}-10i_L$ 代入②化简得

$\frac{d^2i_L}{dt^2}+7\frac{di_L}{dt}+12i_L=12$

特征方程为 $s^2+7s+12=0$ $s_1=-3$ $s_2=-4$

特解为 $i_L(t)=A_1e^{-3t}+A_2e^{-4t}+1$

$u_C(t)=-4A_1e^{-3t}-2A_2e^{-4t}+2$

$\begin{cases}-4A_1-2A_2+2=1\\ A_1+A_2+1=0.5\end{cases}\Rightarrow\begin{cases}A_1=1\\ A_2=-\frac{3}{2}\end{cases}$

故 $u_C(t)=-4e^{-3t}+3e^{-4t}+2$ (V)

$I_L(t)=e^{-3t}-\frac{3}{2}e^{-4t}+1$ (A)

8-27 $i_L(0+)=i_L(0^-)=\frac{E}{R}=0.75A$

$u_C(0+)=u_C(0^-)=5V$

$u_C=0.4\frac{di_L}{dt}+20i_L$

$i_L=-i_C=-0.004\frac{du_C}{dt}$

$\Rightarrow\frac{1}{625}\frac{d^2u_C}{dt^2}+\frac{2}{25}\frac{du_C}{dt}+u_C=0$

$\Rightarrow s_1=s_2=-25$

续8-27

$u_C(t)=(A_1+A_2t)e^{-25t}$

$u_C(0)=A_1=5V$

而 $i_L(t)=-0.004[(A_2-25A_1)e^{-25t}-25A_2te^{-25t}]$

$i_L(0)=0.75A$

$\Rightarrow A_2=-62.5$

$\therefore u_C(t)=(5-62.5t)e^{-25t}$ (V)

8-28

同8-27列方程可知(步骤相同,仅写出特征方程)

① $R=10$ 时

$\frac{1}{625}\frac{d^2u_C}{dt^2}+\frac{1}{25}\frac{du_C}{dt}+u_C=0$ $s_1=-12.5+j1.65$ $s_2=-12.5-j21.65$

故 $u_C(t)=Ae^{-12.5t}\sin(21.65t+\theta)$

而 $i_L(0+)=i_L(0^-)=\frac{5}{10}=0.5A$

$u_C(0+)=u_C(0^-)=5V$

同上代入得 $A=15.3$ $\theta=161°$

即 $u_C(t)=15.3e^{-12.5t}\sin(21.65t+161°)$ V

② $R=30\Omega$ 时

$\frac{1}{625}\frac{d^2u_C}{dt^2}+\frac{3}{25}\frac{du_C}{dt}+u_C=0$ $s_1=-9.549$ $s_2=-65.451$

$u_C(t)=A_1e^{-9.549t}+A_2e^{-65.451t}$

$i_L(t)=-0.004[-9.549A_1e^{-9.549t}-65.451A_2e^{-65.451t}]$

即 $\begin{cases}u_C(0+)=u_C(0^-)=5V\\ I_L(0+)=I_L(0^-)=0.5A\end{cases}\Rightarrow\begin{cases}A_1+A_2=5V\\ 0.0382A_1+0.2618A_2=0.5A\end{cases}$

$\Rightarrow A_1=3.656$ $A_2=1.344$

故 $u_C(t)=3.656e^{-9.549t}+1.344e^{-65.451t}$ (V)

8-29

运用电流法解决问题(同8-23)

将电容看作短路,电感看作开路

即在 $0^-\sim0^+$ 的时间内

$i_C(t)=\frac{\delta(t)}{R}=5\delta(t)$ $u_L(t)=\delta(t)$

$u_C(0+)=u_C(0^-)+\frac{1}{C}\int5\delta(t)dt=5V$ ①

$I_L(0+)=i_L(0^-)+\frac{1}{L}\int\delta(t)dt=6A$ ②

$t>0$ 时

$u_C+(C\frac{du_C}{dt}-i_L)R=0$

$u_C+L\frac{di_L}{dt}=0$ $\Rightarrow\frac{1}{6}\frac{d^2i_L}{dt^2}+\frac{5}{6}\frac{di_L}{dt}+i_L=0$

$\Rightarrow i_L(t)=A_1e^{-2t}+A_2e^{-3t}$ (A) $s_1=-2$ $s_2=-3$

$u_C(t)=-\frac{1}{6}\frac{di_L}{dt}=-\frac{1}{6}(-2A_1e^{-2t}-3A_2e^{-3t})$ (V)

代入①②得 $\begin{cases}A_1+A_2=6\\ \frac{1}{3}A_1+\frac{1}{2}A_2=5\end{cases}\Rightarrow\begin{cases}A_1=-12\\ A_2=18\end{cases}$

即 $i_L(t)=-12e^{-2t}+18e^{-3t}$ (A)

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8-30

$i_L(0^+)=i_L(0^-)=I_S=0.5A$

$u_C(0^+)=u_C(0^-)=R_2I_S=15V$

电路换路后、双一阶电路. 三要素法

左侧 $u_C(\infty)=30V$ $\tau=R_1C=\frac{1}{100}s$

$u_C(t)=30-15e^{-100t}V$

$i_C(t)=C\frac{du_C}{dt}=0.15e^{-100t}A$

右侧 $i_L(\infty)=1A$ $R_{eq}=20\Omega=R_2\|R_3$

$\tau=\frac{L}{R_{eq}}=\frac{1}{200}s$

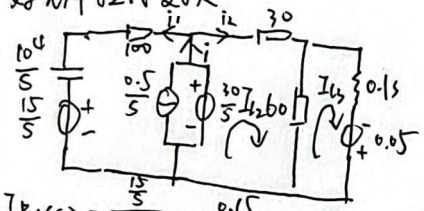
$i_L(t)=1-0.5e^{-200t}$

$i_{R_2}(t)=i_C(t)+\frac{u_L(t)}{R_3}=1-\frac{1}{3}e^{-200t}A$

KCL: $i=i_{R_2}+i_C=1+0.15e^{-100t}-0.33e^{-200t}A$

(答案是0.5. 感觉有点问题. 我用另一种验证也是1)

② 对元件拉氏变换



$I_{R1}(s)=\frac{\frac{10^4}{s}+\infty}{s+100}=\frac{0.15}{s+100}\Rightarrow I_{R1}(t)=0.15e^{-100t}A$

$\begin{cases}\frac{30}{s}=I_{L2}\cdot 30+(I_{L2}-I_{L3})60\\ 0.05=60(I_{L3}-I_{L2})+0.1s\cdot I_{L3}\end{cases}$

$\Rightarrow I_{L2}=\frac{1}{5}-\frac{1}{5}\cdot\frac{1}{s+200}=I_{R2}$

$I_{R2}(t)=1-\frac{1}{3}e^{-200t}$

$i=i_{R2}+i_C=1+0.15e^{-100t}-0.33e^{-200t}A$

8-31

断开前: 节点电压法

$u_a(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3})=\frac{u_S}{R_3}+I_S$

$\Rightarrow u_a=\frac{54}{7}V$

$u_C(0^+)=u_C(0^-)=\frac{54}{7}V$

$i_L(0^+)=i_L(0^-)=\frac{27}{7}A$

断开后

左侧: $i_L(\infty)=3\times\frac{6}{2+6}=2.25A$

$\tau'=\frac{L}{R_{eq}}=\frac{0.1}{R_1+R_2}=\frac{1}{80}s$

$i_L(t)=2.25+1.61e^{-80t}A$

右侧: $u_C(\infty)=u_S=12V$

$\tau''=R_3C=0.2s$

$u_C(t)=12+(-4.286)e^{-5t}V$.

8-32

$u_{C1}(0^+)=u_{C2}(0^+)=RI_S=10V=u_{C1}(0^-)=u_{C2}(0^-)$

断开后 电路结构相同

即 $I_{C1}=C_1\frac{du_{C1}}{dt}=I_{C2}=0.5A$.

$u_{C1}(t)=0.5t+k(V)$

而 $u_{C1}(0)=10V$ 即 $u_{C1}(t)=0.5t+10(V)=u_{C2}(t)$

$u(t)=u_{C2}(t)-R\cdot\frac{1}{2}I_S=0.5t+5(V)$

8-33

$I_{L1}(0^+)=I_{L2}(0^+)=\frac{u_S}{R_1}=1A$

$u_L=L\frac{di_L}{dt}=\frac{u_S}{2}=\frac{1}{2}V$ $i_L=\frac{1}{2}t+k(A)$

$i_{L1}(t)=i_{L2}(t)=\frac{1}{2}t+1(A)$

$i_R=\frac{1}{2}A$

$i_K=i_L-i_R=\frac{1}{2}t+\frac{1}{2}(A)$

8-34 (u_C突变)

$t>0$, $i_L(0^+)=i_L(0^-)=\frac{u_S}{R_1+R_2}=2.5A$

$i_L(\infty)=\frac{u_S}{R_1}=5A$ $\tau_1=\frac{L}{R_{eq}}=\frac{L}{R_1}=1s$

$i_L(t)=5-2.5e^{-t}(A)$

~~$t\le 0$时~~ $u_C(t)=ri_L(t)=25-12.5e^{-t}(V)$

$i_C(t)=C\frac{du_C}{dt}=25e^{-t}(A)$

而$t<0$时 $u_C(0^-)=R_2i_L(0^-)+ri_L(0^-)=17.5V\neq u_C(0^+)$

电压发生突变

则全时间范围内 $(-\infty,+\infty)$内

$u_C(t)=17.5\varepsilon(-t)+[25-12.5e^{-t}]\varepsilon(t)V$

$i_C(t)=-10\delta(t)+25e^{-t}(A)$

$i_K(t)=i_C(t)-i_L(t)=(-5-10\delta(t)+27.5e^{-t})A$

8-35

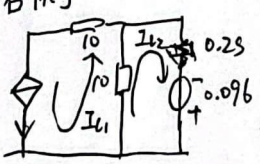
$t<0$时 $i(0^-)=\frac{u_S}{R+(1-p)R}=0.72A$ $i_L(0^-)=0.48A$

$u_C(0^-)=u_S-i(0^-)\cdot R=9.8V$

$t>0$时 左侧 $u_C(\infty)=u_S=12V$ $\tau=RC=0.1s$

$u_C(t)=12-7.2e^{-10t}(V)$ $i_C(t)=C\frac{du_C(t)}{dt}=0.72e^{-10t}(A)$

右侧



$\frac{0.24}{s+10}$

$I_{L1}=\frac{0.24}{s+10}$ $I_{L2}=\frac{\frac{2.4}{s+10}+0.096}{10+0.2s}$

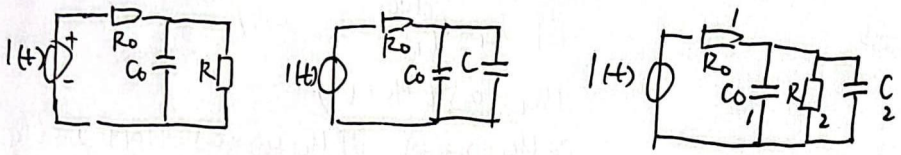
$I_{L2}=\frac{0.78}{s+50}-0.3\cdot\frac{1}{s+10}$

$I_L=I_{L2}(t)=0.78e^{-50t}-0.3e^{-10t}$

$u_K=u_C-[-\frac{1}{3}R+L\frac{di_L}{dt}]=12+7.8e^{-50t}-5.4e^{-10t}V$

8-36

观察 $g_u(t)$ 的形式可知 N_0 的内部结构和作法



$\tau_1 = \frac{2}{3}S \Rightarrow C_0(R\|R_0) = C_0 \cdot \frac{RR_0}{R+R_0} = \frac{2}{3}$ ①

$\tau_2 = 3S \Rightarrow (C_0+C)\cdot R_0 = 3$ ②

而 $t\to\infty$ 时 $u_1(\infty) = \frac{R}{R+R_0} = \frac{2}{3}$ ③ $u_2(\infty) = 1$

由 ① ② ③ $R_0 = 1\Omega$ $C_0 = 1F$ $R = 2\Omega$

$u(\infty) = \frac{2}{3}V$ $\tau = (C_0+C)\cdot\frac{R\cdot R_0}{R+R_0} = 2$

$u = \frac{2}{3}(1-e^{-\frac{t}{2}})V$

9-1

(1) $F(s)=\frac{s+4}{(2s+3)(s+1)}=\frac{K_1}{2s+3}+\frac{K_2}{s+1}$

$K_1=\frac{s+4}{s+1}\Big|_{s=-\frac{3}{2}}=-5$

$K_2=\frac{s+4}{2s+3}\Big|_{s=-1}=3$

$\therefore F(s)=\frac{-5}{2s+3}+\frac{3}{s+1}$ $f(t)=-2.5e^{-1.5t}+3e^{-t}$

(2) $F(s)=\frac{s^2+3s+7}{[(s+1)^2+4](s+1)}=\frac{\dot{K}_{11}}{s+(2-2j)}+\frac{\dot{K}_{12}}{s+(2+2j)}+\frac{K_1}{s+1}$

$K_1=\frac{s^2+3s+7}{(s+1)^2+4}\Big|_{s=-1}=1$

$\dot{K}_{11}=\frac{s^2+3s+7}{(s+2+2j)(s+1)}\Big|_{s=2j-2}=\frac{1}{4}\angle 90^\circ$

而 $f_1(t)=2|K|e^{-\alpha t}\cos(\omega t+\theta)$

本题中 $K=\frac{1}{4}$ $\theta=90^\circ$ $\alpha=2$ $\omega=2$

代入得 $f(t)=e^{-t}+\frac{1}{2}e^{-2t}\cos(2t+90^\circ)$

(3) $F(s)=\frac{2s^2+3s+2}{(s+1)^3}=\frac{K_{13}}{(s+1)^3}+\frac{K_{12}}{(s+1)^2}+\frac{K_{11}}{s+1}$

$K_{13}=(2s^2+3s+2)\Big|_{s=-1}=1$

$K_{12}=\frac{d(2s^2+3s+2)}{ds}\Big|_{s=-1}=-1$

$K_{11}=\frac{1}{2}\frac{d^2(2s^2+3s+2)}{ds^2}\Big|_{s=-1}=2$

即 $F(s)=\frac{2}{s+1}-\frac{1}{(s+1)^2}+\frac{1}{(s+1)^3}$

$f(t)=\frac{1}{2}t^2e^{-t}-te^{-t}+2e^{-t}$

(4) $F(s)=\frac{s+2}{s(s+1)^2(s+3)}=\frac{K_1}{s}+\frac{K_{21}}{(s+1)^2}+\frac{K_{22}}{s+1}+\frac{K_3}{s+3}$

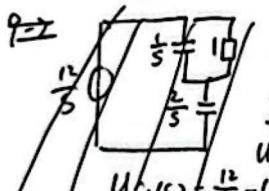
$K_1=\frac{s+2}{(s+1)^2(s+3)}\Big|_{s=0}=\frac{2}{3}$ $K_{21}=\frac{s+2}{s(s+3)}\Big|_{s=-1}=-\frac{1}{2}$

$K_{22}=\frac{d(\frac{s+2}{s(s+3)})}{ds}\Big|_{s=-1}=-\frac{3}{4}$ $K_3=\frac{s+2}{s(s+1)^2}\Big|_{s=-3}=\frac{1}{12}$

即 $F(s)=\frac{\frac{2}{3}}{s}-\frac{\frac{1}{2}}{(s+1)^2}+\frac{-\frac{3}{4}}{s+1}+\frac{\frac{1}{12}}{s+3}$

$f(t)=\frac{2}{3}-\frac{1}{2}te^{-t}-\frac{3}{4}e^{-t}+\frac{1}{12}e^{-3t}$

9-2



$Z_{等}=\frac{1}{25}+\frac{1}{5}\|1=\frac{3s+1}{25(s+1)}$

$I_{等}(s)=\frac{\frac{12}{5}}{Z_{等}}=\frac{24(s+1)}{3s+1}$

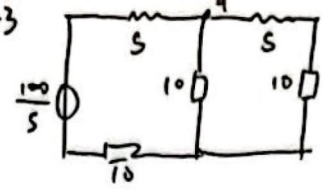
$U_{c2}(s)=\frac{1}{25}\cdot\frac{24(s+1)}{3s+1}=\frac{K_1}{s}+\frac{K_2}{3s+1}$

$K_1=\frac{24(s+1)}{2(3s+1)}\Big|_{s=0}=12$ $K_2=\frac{24(s+1)}{2s(3s+1)}\Big|_{s=-\frac{1}{3}}=-24$

$U_{c2}(t)=12-8e^{-\frac{1}{3}t}(V)$

$U_{c1}(t)=E-U_{c2}(t)=8^{-\frac{1}{3}t}(V)$

9-3



对a点列节点电压

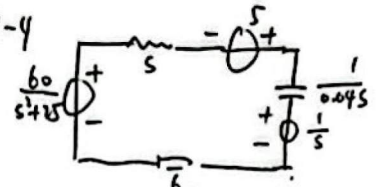
$U_a(\frac{1}{10}+\frac{1}{5+10}+\frac{1}{5+10})=\frac{\frac{100}{s}}{s+10}$

$\Rightarrow U_a=\frac{100}{s(s+30)}$

$I_2(s)=\frac{U_a}{s+10}=\frac{1000}{s(s+10)(s+30)}=\frac{\frac{10}{3}}{s}+\frac{-5}{s+10}+\frac{\frac{5}{3}}{s+30}$

$I_2(t)=\frac{10}{3}-5e^{-10t}+\frac{5}{3}e^{-30t}$

9-4

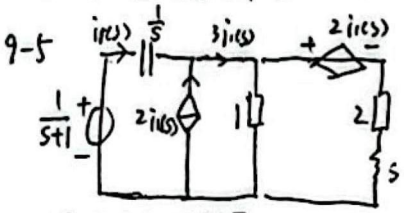


$I_L(s)=\frac{\frac{60}{s+8}+5-\frac{1}{s}}{s+\frac{1}{0.04s}+6}=\frac{60s+(5s-1)(s^2+8s)}{(s^2+8s)(s^2+6s+25)}$

$=\frac{1\angle 90^\circ}{s+5j}+\frac{1\angle -90^\circ}{s-5j}+\frac{4.1\angle 52.9^\circ}{s+3+4j}+\frac{4.1\angle -52.9^\circ}{s+3-4j}$

$i_L(t)=2\cos(5t-90^\circ)+8.2e^{-3t}\cos(4t+52.9^\circ)A$

9-5



有两个KVL方程

$\begin{cases}\frac{1}{s+1}=\frac{1}{s}\cdot i_1(s)+[\beta i(s)-i_{L1}(s)]\cdot 1\\ 0=[i_{L1}-3i_1(s)]+2i_1(s)+(k+s)i_{L1}\end{cases}$

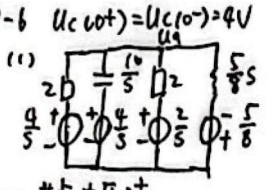
$\Rightarrow i_{L1}(s)=\frac{s}{(s+1)(3s^2+9s+3)}$

$=\frac{\frac{1}{3}}{s+1}+\frac{-0.092}{s+0.382}+\frac{-0.241}{s+2.618}$

$i(t)=\frac{1}{3}e^{-t}-0.092e^{-0.382t}-0.241e^{-2.618t}(A)$

9-6 $U_C(0+)=U_C(0-)=4V$ $I_L(0+)=I_L(0-)=\frac{U_{S2}}{R}=1A$

(1)



(2) 节点电压法

$U_a(\frac{1}{2}+\frac{5}{10}+\frac{1}{2}+\frac{8}{5s})=\frac{2}{5}+\frac{2}{5}+\frac{1}{5}-\frac{1}{3}$

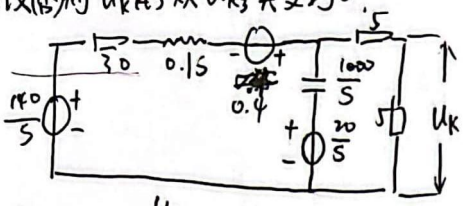
$\Rightarrow U_a(s)=\frac{10(10+2s)}{5s^2+50s+80}=\frac{2}{s+2}+\frac{2}{s+8}$

(3) $U_a(t)=2e^{-2t}+2e^{-8t}(V)$

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9-7 (我验算了一下，书后面答案是用下闭合做的)

不改图则 $u_K(t)$ 从 u_{R_3} 突变到0.



$u_C(0_-) = \frac{u_S}{R_1+R_2}\cdot R_2 = 20V = u_C(0_+)$

$i_L(0^-) = \frac{u_S}{R_1+R_2} = 4A$

由节点电压法

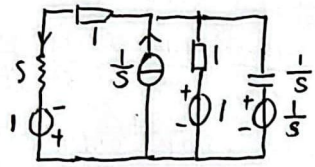
$$\left(\frac{1}{30+0.1s}+\frac{s}{1000}+\frac{1}{10}\right)U_n(s) = \frac{\frac{140}{s}+40i}{30+0.1s}+\frac{\frac{20}{s}}{\frac{1000}{s}}$$

$$\Rightarrow U_n(s) = \frac{20(s^2+300s+7000)}{s(s+20)^2} = \frac{35}{s}+\frac{-1000}{(s+200)^2}+\frac{-15}{s+200}$$

而 $U_K = \frac{1}{2}U_n$

$$\therefore u_K(t) = \frac{1}{2}\left[35-1000te^{-200t}-15e^{-200t}\right]$$

9-8 $i_L(0^-) = I_S = 1A \quad u_C(0^-) = R_1 I_S = 1V$



列节点电压方程

$$\left(\frac{1}{s+1}+1+s\right)U_C(s) = -\frac{1}{s+1}+\frac{1}{s}+1+1$$

$$\Rightarrow U_C(s) = \frac{2s^2+2s+1}{s(s^2+2s+2)}$$

9-9 $H(s) = \frac{s+1}{s^2+5s+6} = \frac{-1}{s+2}+\frac{2}{s+3}$

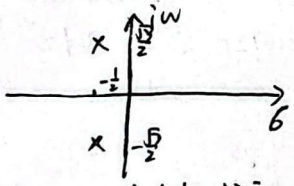
$h(t) = -e^{-2t}+2e^{-3t}$

$R(s) = H(s)E(s) = \frac{s+1}{s(s^2+5s+6)} = \frac{\frac{1}{6}}{s}+\frac{\frac{1}{2}}{s+2}+\frac{-\frac{2}{3}}{s+3}$

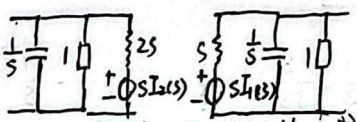
$r(t) = \frac{1}{6}+\frac{1}{2}e^{-2t}-\frac{2}{3}e^{-3t}$

9-10 $\frac{U_2(s)}{U_1(s)} = \frac{\frac{1}{s}/\!/1}{1+\frac{1}{s}/\!/1} = \frac{1}{s^2+s+1}$

$s_1 = -\frac{1}{2}+j\frac{\sqrt{3}}{2} \quad s_2 = -\frac{1}{2}-j\frac{\sqrt{3}}{2}$



9-11 (本题中有互感的频域变化方式)



互感用流控电压源代替互感

$$\begin{cases} U_2(s)\left(\frac{1}{s}+s+1\right) = \frac{sI_1(s)}{s} \\ U_1(s) = 2sI_1(s)+sI_2(s) \\ U_2(s) = sI_2(s)+sI_1(s) \end{cases}$$

$$\Rightarrow U_2(s) = \frac{1}{s^2+s+1}(U_1(s)-U_2(s))$$

即 ~~$H(s)$~~ $H(s) = \frac{U_2(s)}{U_1(s)} = \frac{1}{s^2+s+2}$

9-12

$H(s) = \frac{H_0}{s^2+2s+2}$ 可画幅频曲线

$H(j\omega) = \frac{H_0}{(j\omega)^2+2j\omega+2}$ 可画相频曲线

9-13

(1) $H(s) = \frac{100}{s+200} \quad H(j\omega) = \frac{100}{j\omega+200}$

(2) $R(s) = H(s)E(s) = \frac{100}{s(s+200)} = \frac{\frac{1}{2}}{s}-\frac{\frac{1}{2}}{s+200}$

~~$R(t)$~~ ~~$r(t) = \frac{1}{2}e$~~

$r(t) = \left(\frac{1}{2}-\frac{1}{2}e^{-200t}\right)\varepsilon(t)$

9-14

输入为 $\varepsilon(t)$ 时

$u_C(s) = \frac{1}{s}\cdot\frac{1}{1+\frac{1}{s}}\cdot\frac{1}{s} = \frac{1}{s}-\frac{1}{s+1}$

$u_C(t) = 1-e^{-t}(V)$

输入为 $\delta(t)$ 时

$u_C(s) = \frac{1}{1+\frac{1}{s}}\cdot\frac{1}{s} = \frac{1}{s+1} \quad u_C(t) = e^{-t}(V)$

而题目中，输入为 $1(t)-1(t-T_1)+\delta(t-T_2)+\delta(t-T_3)$

故输出为

$u_C(t) = \frac{1}{2}(1-e^{-t})\cdot 1(t)-(1-e^{-t+T_1})1(t-T_1)$

$+e^{-(t-T_2)}\varepsilon(t-T_2)+e^{-t-T_3}\varepsilon(t-T_3)$

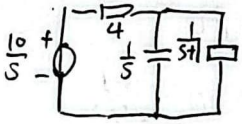
9-15 (1象)

找到 N 网络在频域中等效电阻

① 未串 R、C 时

$R(s) = \frac{1}{s}-\frac{1}{s+1} \quad E(s) = \frac{1}{s} \quad H(s) = \frac{R(s)}{E(s)} = \frac{1}{s+1}$

② 串 R、C 时，电路图如下



$I(s) = \frac{\frac{10}{s}}{4+\frac{1}{s}/\!/\frac{1}{s+1}} = \frac{2}{s}+\frac{4}{8s+5}$

$i(t) = 2+\frac{4}{8}e^{-\frac{5}{8}t}$

$= 2+\frac{1}{2}e^{-\frac{5}{8}t}(A)$

9-16

$H(s) = \frac{1}{s+2}$

故先求零状态响应

$I_1(s) = \frac{10}{s}\cdot\frac{1}{s+2} = \frac{5}{s}-\frac{5}{s+2}$

$i_1(t) = 5-5e^{-2t}(A)$

零输入响应 $i_2(t) = 2e^{-2t}(A)$

$i = i_1(t)+i_2(t) = 5-3e^{-2t}(A)$

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9-17

(1) $\frac{U_2(s)}{U_1(s)} = \frac{A_1}{s+4.5} + \frac{A_2}{s+8}$

由 $u_2(0_+)=50V \Rightarrow A_1+A_2=50$ ①

当 $u_1(t)=1(t)$ 时

$U_2(s) = \frac{A_1}{s(s+4.5)} + \frac{A_2}{(s+8)s}$

$= \frac{K_1}{s} + \frac{K_2}{s+4.5} + \frac{K_3}{s+8}$

求稳态响应电压，求 K_1 即可（或用终值定理）

$K_1 = \frac{A_1}{4.5} + \frac{A_2}{8} = 0$ ②

由①②可知 $A_1 = -\frac{450}{7}$ $A_2 = \frac{800}{7}$

$\therefore H(s) = \frac{U_2(s)}{U_1(s)} = \frac{-\frac{450}{7}}{s+4.5} + \frac{\frac{800}{7}}{s+8} = \frac{50s}{(s+4.5)(s+8)}$

(2) $U_1(s) = \frac{10\frac{1}{2}\cdot\sin\theta + w\cos\theta}{s^2+w^2} = \frac{\frac{1}{2}s+3\sqrt{3}}{s^2+36}$

$U_2(s) = \frac{10(\frac{1}{2}s+3\sqrt{3})}{s^2+36} \cdot \frac{50s}{(s+4.5)(s+8)}$

$= \frac{K_1}{s+j6} + \frac{K_2}{s-j6} + \frac{K_3}{s+4.5} + \frac{K_4}{s+8}$

求 K_1, K_2 即可 $K_1 = 20\angle 60°$ $K_2 = 20\angle -60°$

即稳态电压 $u_{2N}(t) = 40\cos(6t-60°)V$

9-18

对图a I_S 右侧作戴维南等效

$u_S(t) = u_L(t) + u_3(t)$

$= 2+2e^{-4t}(V)$

$U_S(s) = \frac{2}{s} + \frac{2}{s+4}$ $I_S(s) = \frac{2}{s}$

等效电阻 $R_S = \frac{U_S(s)}{I_S(s)} = \frac{2s+4}{s+4}$

对图b 用互易定理（第三种特殊情况）P57

$I_3(s) = \frac{U_3(s)}{6} = \frac{\frac{1}{3}}{s} - \frac{\frac{1}{3}}{s+4}$

$\frac{I_3(s)}{I_S(s)} = \frac{U_{OC}(s)}{U_S(s)} \Rightarrow U_{OC}(s) = \frac{\frac{2}{3}}{s(s+4)}$

$I'_S(s) = \frac{U_{OC}(s)}{R_S+2} = \frac{\frac{1}{18}}{s} - \frac{\frac{1}{18}}{s+4}$

$I'_S(t) = (\frac{1}{18} - \frac{1}{18}e^{-4t})A$

9-19 （答案错误，应为 $(-\frac{5}{18}e^{-\frac{1}{2}t} + \frac{25}{9}e^{-5t})\cdot 1(t)A$）

① 方法1：（当然也可以用2参数2方程）

考虑到为无受控源 or 独立源双口网络

我们采用T型等效电路 N ⇔ z_{11} z_{22} z_{12}

图a：$i_1(s)$ z_{11} z_{12} z_{22} $U_{oc}(s)$

$i_1(s) = \frac{1}{5s+1}$ $U_{oc}(s) = 1 - \frac{1}{5s+1} = \frac{5}{5s+1}$

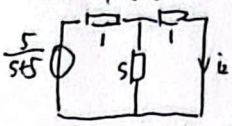
$\Rightarrow z_{12} = 5$ $z_{11} = 1$

图b：z_{11} z_{22} $5s+1$ z_{12} $i_{2a}(s)$

$I_{2a}(s) = 1 - \frac{1}{5s+1} = \frac{5}{5s+1} = 1 \times \frac{z_{12}}{z_{12}+z_{22}}$

$\Rightarrow z_{22} = 1$

故图c中可等效成



$I_2(s) = \left[\frac{5}{5s+5} \Big/ (1+1//5)\right] \cdot \frac{5}{5s+1}$

$= \frac{5s}{(2s+1)(s+5)} = \frac{-\frac{5}{9}}{2s+1} + \frac{\frac{25}{9}}{s+5}$

$i_2(t) = -\frac{5}{18}e^{-\frac{1}{2}t} + \frac{25}{9}e^{-5t}$

② 运用特勒根定理求解

先对图a与图b列特勒根（互易定理）

注意方向：

$-U_{1b}(s)I_{1a}(s) - U_{1a}(s)I_{2b}(s) + U_{2a}(s)I_{2b}(s) = -U_{1a}(s)I_{1b}(s) + U_{2b}(s)I_{2a}(s)$

$U_{1a}(s) = 1$ $U_{2a}(s) = 1 - \frac{1}{5s+1} = \frac{5}{5s+1}$ $I_{1a}(s) = \frac{1}{5}$ $I_{2a}(s) = 0$

$U_{1b}(s)$ 未知 $U_{2b}(s) = 0$ $I_{1b}(s) = -1$ $I_{2b}(s) = \frac{5}{5s+1}$

代入可得 $U_{1b}(s) = \frac{2s+1}{s+1}$

对于图b与图c，有

$\frac{U_{1b}(s)}{I_{2b}(s)} = \frac{U_{1c}(s)}{I_{2c}(s)} \Rightarrow \frac{\frac{2s+1}{s+1}}{\frac{5}{5s+1}} = \frac{5}{I_2(s)}$

$\Rightarrow I_2(s) = \frac{\frac{25}{9}}{s+5} - \frac{\frac{5}{18}}{s+\frac{1}{2}}$

即 $I_2(t) = \frac{25}{9}e^{-5t} - \frac{5}{18}e^{-\frac{1}{2}t}$

9-20 同上，用互易定理或T型等效。

T型等效

$1 \Leftrightarrow \delta(t)$ z_{11} z_{22} z_{12} U_2

$U_2(s) = \frac{z_{12}}{z_{11}+z_{12}} = \frac{2}{s+1} + \frac{3}{s+2}$

假设用 $1(t)$ 激励

i_1 z_{11} z_{22} z_{12} $\frac{1}{5}$

$I_1(s) = \frac{1}{5}\cdot\frac{z_{11}}{z_{11}+z_{12}} = \frac{1}{5}(\frac{2}{s+1} + \frac{3}{s+2}) = \frac{\frac{2}{5}}{s} + \frac{-2}{s+1} + \frac{-\frac{3}{5}}{s+2}$

改用 $1(t-0.1)$ 激励

$I_1(t) = [\frac{2}{5} - 2e^{-(t-0.1)} - \frac{3}{5}e^{-2(t-0.1)}]1(t-0.1)A$

9-21 N为有源网络，故分零状态响应和零输入响应

(1) $u_S = 5e^{-2t}$ 时

零输入响应 $i_{2i}(t) = 2(1+3e^{-2t}-5e^{-t}) - (1+6e^{-2t}-8e^{-t})$

$= 1-2e^{-t}(A)$

零状态响应 $i_{2s}(t) = i_2(t) - i_{2i}(t) = 3e^{-2t} - 3e^{-t}(A)$

$u_S = 20e^{-2t}$ 时

零状态响应 $i'_{2s}(t) = 4i_{2s}(t) = 12e^{-2t} - 12e^{-t}(A)$

$i'_2(t) = i'_{2s}(t) + i_{2i}(t) = 1+12e^{-2t}-14e^{-t}(A)$

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9-21 课

(2) $U_{1S}(s) = 10 \cdot \frac{2}{s^2+4} = \frac{20}{s^2+4}$

$H_{2S}(s) = \frac{\frac{3}{s+2} - \frac{3}{s+1}}{\frac{5}{s+2}} = -\frac{3}{5} \cdot \frac{1}{s+1}$

$I_{2S}(s) = U_{1S}(s) \cdot H(s) = \frac{-12}{(s+1)(s^2+4)}$

$= \frac{-2.4}{s+1} + \frac{1.34\angle 116.56^\circ}{s-2j} + \frac{1.34\angle -116.56^\circ}{s+2j}$

即 $i_{2S}(t) = (-2.4e^{-t} + 2.68\cos(2t+26.56^\circ))$ A

$i_2(t) = i_{2S}(t) + i_{2i}(t) = (-4.4e^{-t} + 2.68\cos(2t+26.56^\circ))$ A

(与答案不一致，感觉我的是对的)

9-22

(1) 同 9-21

$U_1(s) = 1 \quad U_{2i}(s) = \frac{2}{s+1}$

$U_1(s) = \frac{1}{s} \quad U_{2s}(s) = \frac{3}{s} + \frac{4}{s+1}$

即 ~~$\frac{2}{s+1} = U_{2i}(s) + U_{2s}(s)$~~

~~$\frac{3}{s} + \frac{4}{s+1} = U_{2i}(s) + \frac{1}{s}U_{2s}(s)$~~

$\frac{2}{s+1} = U_{2i}(s) + H_s(s)$

$\frac{3}{s} + \frac{4}{s+1} = U_{2i}(s) + \frac{1}{s}H_s(s)$

$\Rightarrow H_s(s) = \frac{5s+3}{(s-1)(s+1)} \quad U_{2i}(s) = \frac{7s+1}{(s-1)(s+1)} = \frac{4}{s-1} + \frac{3}{s+1}$

即 U_2 的零输入响应 $U_{2zi}(t) = 4e^{t} + 3e^{-t}$ (V)

(2) $H(s) = \frac{5s+3}{(s-1)(s+1)} = -\frac{1}{s+1} - \frac{4}{s-1}$ $(h(t) = -e^{-t} - 4e^{t})$

~~① $U_2'(s) = U_{2i}(s) + H(s)$~~

~~② $U_2''(s) = U_{2i}(s) + \frac{1}{s}H(s)$~~

(3) ① 输入为 $1(t)$ 时

$U_{2S}(s) = \frac{2}{s}H(s) + U_{2i}(s) = \frac{5}{s+1} - \frac{4}{s-1} + \frac{6}{s}$

② 输入为 $1(t-t_0)$ 时

$U_{2S}(s) = \frac{2}{s}H(s) = \frac{2}{s}(-\frac{1}{s+1} - \frac{4}{s-1}) = \frac{6}{s} - \frac{8}{s-1} + \frac{2}{s+1}$

③ 当输入为 $\delta(t-2t_0)$ 时

$U_{2S}(s) = H(s) = -\frac{1}{s+1} - \frac{4}{s-1}$

组合起来：

即 $U_2(t) = [6 + 5e^{-t} - 4e^{t}) \cdot 1(t) - (6 + 2e^{-(t-t_0)} - 8e^{(t-t_0)}) \cdot 1(t-t_0)$

$+ [e^{-(t-2t_0)} - 4e^{(t-2t_0)}] \cdot 1(t-2t_0)$

9-23

(1) 由 $H(s)$ 的零极点可知 $H(s) = \frac{K(s+2)}{s^2+6s+13}$

而 $E(s) = \frac{26}{s}$

故 $U_2(s) = \frac{26K(s+2)}{s(s^2+6s+13)}$

$\lim_{t\to\infty} U_2(t) = \lim_{s\to 0} sU_2(s) = \lim_{s\to 0} \frac{26K(s+2)}{s^2+6s+13} = 8 \Rightarrow K = 2$

$\therefore H(s) = \frac{2(s+2)}{s^2+6s+13}$

课 9-23

(2) $E(s) = 141.4 \times \frac{\frac{1}{2}s + \frac{\sqrt{3}}{2} \times 2}{s^2+4}$

$U_2(s) = H(s) \cdot E(s) = 141.4 \times \frac{\frac{1}{2}s + \sqrt{3}}{s^2+4} \times \frac{2(s+2)}{s^2+6s+13}$

$= \frac{K_1}{s+2j} + \frac{K_2}{s-j2} + \cdots\cdots$

求稳态响应 故只需求 K_1, K_2

$K_1 = 26.66\angle 68.13^\circ \quad K_2 = 26.66\angle -68.13^\circ$

故 $U_{2p}(s) = 53.32\cos(2t - 68.13^\circ)$

9-24

$y(t) = \int_0^{t_0} e_1(\tau)h(t-\tau)d\tau + \int_{t_0}^{t} e_2(\tau)h(t-\tau)d\tau$

9-25 P289页 9-9-14

$y(t) = e_1(0)g(t) + \int_0^{t_0} e_1'(\tau)g(t-\tau)d\tau + \int_{t_0}^{t} e_2'(\tau)g(t-\tau)d\tau$

$- [e_1(t_0) - e_2(t_0)]g(t-t_0)$

9-26 (裴阿梅尔积分不做要求) 只做第一问

$H(s) = \frac{I(s)}{U(s)} = \frac{1}{s+1}$ 即 $h(t) = e^{-t}$

即用 $y(t) = \int_0^t e(\tau)h(t-\tau)d\tau$ 卷积可得

$0 \le t \le 1$ 时

$i(t) = \int_0^t \tau \cdot e^{-(t-\tau)}d\tau = e^{-t} + t - 1$ (A)

$t > 1$ 时

$i(t) = \int_0^1 \tau e^{-(t-\tau)}d\tau = e^{-t}$ (A)

9-27

$H(s)$ 与 9-26 相同

$0 \le t \le 0.2$ 时 $i(t) = \int_0^t 10 \cdot e^{-2\tau} \cdot e^{-(t-\tau)}d\tau = 10e^{-t} + 10e^{-2t}$ (A)

$t > 0.2$ 时 $i(t) = \int_0^{0.2} 10e^{-2\tau}e^{-(t-\tau)}d\tau + \int_{0.2}^{t} 2e^{-(t-\tau)}d\tau$

$= 2 - 0.63e^{-t}$ (A)

9-28

$0 \le t \le 1$ 时 $i(t) = \int_0^t \tau e^{-(t-\tau)}d\tau = e^{-t} + t - 1$ (A)

$t > 1$ 时 $i(t) = \int_0^1 \tau e^{-(t-\tau)}d\tau + \int_1^t e^{-(t-\tau)}d\tau$

$= e^{-t} + 1 - e^{-t+1}$ (A)

9-29

$$\begin{cases} L\frac{di_L}{dt} = U_S - U_C \\ C\frac{dU_C}{dt} = i_L - \frac{U_C}{R} \end{cases} \Rightarrow \begin{bmatrix} \frac{dU_C}{dt} \\ \frac{di_L}{dt} \end{bmatrix} = \begin{bmatrix} -4 & 2 \\ -\frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} U_C \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} U_S$$

9-30

$$\begin{cases} C\frac{dU_C}{dt} = -i_{L_1} - i_{L_2} \\ L_1\frac{di_{L_1}}{dt} = U_C - U_S - R_1 i_{L_1} \\ L_2\frac{di_{L_2}}{dt} = U_C - i_{L_2}R_2 \end{cases} \Rightarrow \begin{bmatrix} \frac{dU_{C1}}{dt} \\ \frac{di_{L1}}{dt} \\ \frac{di_{L2}}{dt} \end{bmatrix} = \begin{bmatrix} 0 & -1 & -1 \\ 1 & -1 & 0 \\ 2 & 0 & -4 \end{bmatrix} \begin{bmatrix} U_C \\ i_{L1} \\ i_{L2} \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} U_S$$

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9-31

$$\begin{cases} L_7\frac{di_{L7}}{dt}=u_{C3}-u_{C2} \\ L_8\frac{di_{L8}}{dt}=u_{C4}+R_5 i_{R5} \\ i_{R5}=i_{S9}-i_{L8} \\ C_2\frac{du_{C2}}{dt}=i_{L7} \\ C_3\frac{du_{C3}}{dt}=-i_{L7}-i_{R6} \\ i_{R6}=\frac{1}{R}(-u_{S1}+u_{C3}+u_{C4}) \\ C_4\frac{du_{C4}}{dt}=-i_{L8}-i_{R6} \end{cases} \Rightarrow \begin{bmatrix} \frac{du_{C2}}{dt} \\ \frac{du_{C3}}{dt} \\ \frac{du_{C4}}{dt} \\ \frac{di_{L7}}{dt} \\ \frac{di_{L8}}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{C_2} & 0 \\ 0 & -\frac{1}{C_3} & -\frac{1}{C_3} & -\frac{1}{C_3} & 0 \\ 0 & -\frac{1}{C_4} & -\frac{1}{C_4} & 0 & -\frac{1}{C_4} \\ -\frac{1}{L_7} & \frac{1}{L_7} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{L_8} & 0 & -\frac{R_5}{L_8} \end{bmatrix} \begin{bmatrix} u_{C2} \\ u_{C3} \\ u_{C4} \\ i_{L7} \\ i_{L8} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{1}{C_3} & 0 \\ \frac{1}{C_4} & 0 \\ 0 & 0 \\ 0 & \frac{R_5}{L_8} \end{bmatrix} \begin{bmatrix} u_9 \\ I_{S9} \end{bmatrix}$$

9-32

$$\begin{cases} C_3\frac{du_{C3}}{dt}=i_{L1}-i_{L2} \\ L_1\frac{di_{L1}}{dt}=u_S-R_5 i_{L1}-u_{C3} \\ L_2\frac{di_{L2}}{dt}=u_{C3}-i_{R4}R_4 \\ i_{R4}=i_{L2}+i_S \end{cases} \Rightarrow \begin{bmatrix} \frac{du_{C3}}{dt} \\ \frac{di_{L1}}{dt} \\ \frac{di_{L2}}{dt} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{C_3} & -\frac{1}{C_3} \\ -\frac{1}{L_1} & -\frac{R_5}{L_1} & 0 \\ \frac{1}{L_2} & 0 & -\frac{R_4}{L_2} \end{bmatrix} \begin{bmatrix} u_{C3} \\ i_{L1} \\ i_{L2} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{1}{L_1} & 0 \\ 0 & -\frac{R_4}{L_2} \end{bmatrix} \begin{bmatrix} u_S \\ I_S \end{bmatrix}$$

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10-1

$z_0 = R_0 + j\omega L_0 = 0.408\angle 78.7°\ \Omega/km$

$Y_0 = G_0 + j\omega C_0 = 2.8\times10^{-6}\angle 90°\ S/km$

$z_c = \sqrt{\frac{z_0}{Y_0}} = 381.7\angle -5.65°$　$\gamma = \sqrt{z_0 Y_0} = 1.07\times10^{-3}\angle 84.35°$

10-2

已知终端电压电流时

$$\begin{cases} \dot U(x') = \dot U_2 \cdot \frac{e^{\gamma x}+e^{-\gamma x}}{2} + \dot I_2 z_c \cdot \frac{e^{\gamma x}-e^{-\gamma x}}{2} & ① \\ \dot I(x') = \dot I_2 \cdot \frac{e^{\gamma x}+e^{-\gamma x}}{2} + \frac{\dot U_2}{z_c}\cdot\frac{e^{\gamma x}-e^{-\gamma x}}{2} & ② \end{cases}$$

而 $\gamma = \sqrt{z_0 Y_0} = 1.07\times10^{-3}\angle 84.35°$　$x = L = 300\times10^3 m = 300km$

$\dot U_2 = 220\angle 0° V = 110\sqrt2\angle 0° kV$　$U_P = \frac{110\sqrt2}{\sqrt3}\angle 0° = 89.8\angle 30°\ kV$

$\dot I_2 = \frac{P}{3U_P\cdot\cos\varphi} = 0.412\ kA$　(注意为三相)

$\dot I_2$ 滞后 $\dot U_2$ $\varphi = 25.84°$ 即 $\dot I_2 = 0.412\angle -55.84°\ kA$

至此 $\dot U_2, \dot I_2, z_c, \gamma, x$ 均已知 (注: $\dot U_2$ 应代入相电压)

代入①、②即可求解得到 $\dot U_1 = 122.12\angle -10.57°\ kV$　$\dot I_1 = 0.366\angle -44.66°\ kA$

$P_1 = 3U_1 I_1\cos\Theta = 3\times122.12\times0.366\cos(10.57-44.66) = 111.046\ MW$

$\eta = \frac{P_2}{P_1}\times100\% = 90.05\%$

10-3　$\dot I_2 = 0$，即 $\dot U_{2P} = \dot U_1 \Big/ \frac{e^{\gamma l}+e^{-\gamma l}}{2} = 128.55\angle -11.17°\ kV$

$\dot U_{2L} = \sqrt3\,\dot U_{2P}\angle 30° = 222.65\angle 18.83°\ kV$

10-4　$\dot U_2 = 0$　$\dot I_2 = \frac{\dot U_S}{z_{in}} = \frac{\dot U_S}{z_c\cdot sh\gamma x} = \frac{\dot U_S}{z_c\cdot\frac{e^{\gamma x}-e^{-\gamma x}}{2}} = 1.672\angle -89.98°\ kA$

10-5?　$\dot U_1 = \dot U_2 e^{\gamma l}$　$\gamma = \beta + j\alpha = 0.8 + j0.5$

即 $\dot U_2 = \frac{\dot U_1}{e^{\gamma l}}$

$\dot I_2 = \frac{\dot U_2}{z_c}$

(跟答案没对上，而且 R_0 给的很迷)

10-6　AB端右侧为无反射线

即 $z_{C2} = R_2 = 700\Omega$　即 AB右端等效电阻 $z_{i2} = 700\Omega$

即 $z_{C1} = \frac{R_1 z_{i2}}{R_1 + z_{i2}} \Rightarrow R_1 = 400\Omega$

10-7

$$\begin{cases} \dot U_1 = \dot U_2 \cdot \frac{e^{\gamma x}+e^{-\gamma x}}{2} + \dot I_2 z_c \cdot \frac{e^{\gamma x}-e^{-\gamma x}}{2} & ① \\ \dot I_1 = \dot I_2 \cdot \frac{e^{\gamma x}+e^{-\gamma x}}{2} + \frac{\dot U_2}{z_c}\cdot\frac{e^{\gamma x}-e^{-\gamma x}}{2} & ② \end{cases}$$

即 $z_{1i} = z_c \cdot \frac{z_2 ch\gamma l + z_c sh\gamma l}{z_2 sh\gamma l + z_c ch\gamma l}$　⇐①/② ③

AB右端为无反射线即 $Z_{AB右} = z_{C2} = R_2 = z_C$

$z_{AB} = \frac{z_{AB右}-R_1}{z_{AB右}+R_1} = \frac{z_c^2}{2z_c} = \frac{z_c}{2}$

代入③得 $z_{1i} = z_c\cdot\frac{ch\gamma l + 2sh\gamma l}{2ch\gamma l + sh\gamma l}$

10-8

匹配时 $z_i = z_c = 300\Omega$

$R = 600\Omega$ 时

$\alpha = \frac{2\pi}{\omega\lambda} = \frac{2\pi}{v/f} = \frac{4}{3}\pi$

$z_i = \frac{\dot U_2\cos\alpha x + j z_c \dot I_2 \sin\alpha x}{\dot I_2\cos\alpha x + j\frac{\dot U_2}{z_c}\sin\alpha x} = z_c\cdot\frac{z_2 + j z_c\tan\alpha x}{z_c + j z_2\tan\alpha x}$

$x = L = 1.5m$　即 $z_i = 600\Omega$

10-9

$$\begin{cases} \lambda = \frac{v}{f} \\ \frac{1}{\omega C} = z_c\cot\alpha L \end{cases} \Rightarrow L = \frac{\lambda}{4} = 0.25m$$

10-10　($z = j\omega L$　$L_1 = \frac{\lambda}{4}$)

$z = j z_c\tan\alpha L_1 \Rightarrow L = 0.54m$

10-11

速度为光速

$\lambda = \frac{v}{f} = 0.4m$

10-12

L_2 段 $z_2' = j z_c\tan\alpha L_2$

L_1 段 $z_1' = z_c\cdot\frac{z_2 + j z_c\tan\alpha L_2}{z_c + j z_2\tan\alpha L_1}$　$\alpha = \frac{2\pi}{\lambda}$

$z_1' \| z_2' = z_c$

$$\Rightarrow \begin{cases} L_1 = 0.56m \\ L_2 = 1.128m \end{cases}$$

10-13

$z_3' = j z_c\tan\alpha L_3 = j260\Omega$

$z_2' = -j z_c\cot\alpha L_2 = j50\sqrt3\Omega$

$z \| z_2' \| z_3' = 150\Omega \Rightarrow z = 59.6\angle -66.57°$

$\dot U_1 = \dot U_2(\cos\alpha L_1 + j\sin\alpha L_1)$

$\Rightarrow \dot U_2 = 10\angle -165°\ V$

$\dot U_{20} = \frac{\dot U_2}{\cos\alpha L_2} \Rightarrow \dot U_{20} = 20\angle 45°\ V$

$\dot U_2 = j z_c \dot I_{3d}\sin\alpha L_3 \Rightarrow \dot I_{3d} = 0.077\angle 105°$

10-14

$\dot U_2$ 右侧等效电阻 $z_2' = \frac{R_2 + j z_{c2}\tan\alpha L_3}{z_{c2} + j R_2\tan\alpha L_3} = 300\Omega = z_{C1}$

$\therefore z_{C1}$ 为无反射线

$\dot U_1$ 右侧等效电阻 $z_1' = 300\Omega$

$\dot U_1 = \frac{\dot U_S}{R_1 + z_1'}\cdot z_1' = 2\angle 0°\ V$

$\dot U_1 = \dot U_2(\cos\alpha L_1 + j\sin\alpha L_1) \Rightarrow \dot U_2 = 2\angle 45°\ V$

$\dot I_2 = \frac{\dot U_2}{z_{C1}} = \frac{1}{150}\angle -45°$　($\alpha L_3 = \frac{\pi}{2}$)

$$\begin{cases} \dot U_2 = \dot U_3\cos\alpha L_3 + j z_{C3}\dot I_3\sin\alpha L_3 \\ \dot I_2 = j\frac{\dot U_3}{z_{C2}}\sin\alpha L_3 + \dot I_3\cos\alpha L_3 \end{cases} \Rightarrow \dot U_3 = \frac{2\sqrt3}{3}\angle -135°\ V$$

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